

1)

```
test> show dbs
ShopDB      72.00 KiB
admin       40.00 KiB
config      84.00 KiB
ecommerce   72.00 KiB
local       104.00 KiB
mydb        212.00 KiB
peopleDB    48.00 KiB
test        8.00 KiB
todo        48.00 KiB
xyz_institute 16.00 KiB
test> use company
switched to db company
company> db
company
company> db.createCollection("employees")
{ ok: 1 }
company>
```

```
company> db.employees.insertOne({ EmployeeID: 101, Name: "John Doe", Department: "HR", Position: "Manager", Salary: 75000})
{
  acknowledged: true,
  insertedId: ObjectId('67736437ef3f907b892710bd')
}
```

```
company> db.employees.find().pretty()
[
  {
    _id: ObjectId('67736437ef3f907b892710bd'),
    EmployeeID: 101,
    Department: 'HR',
    Position: 'Manager',
    Salary: 75000
  }
]
company>
```

```
company> db.employees.drop()
true
company> exit
```

2)

```

test> use corporate_hub
switched to db corporate_hub
corporate_hub> db.employees.insertOne({ EmployeeID: 2001, Name: "Michael Brown", Position: "Software Engineer", Salary: 85000, DateofJoining: "20020-06-15"})
{
  acknowledged: true,
  insertedId: ObjectId('677365f813de0e7bb72710bc')
}
corporate_hub> db.epmloyees.find({}, {Name:1, Position: 1, _id: 0})

corporate_hub> db.employees.find({}, {Name:1, Position: 1, _id: 0})
[ { Name: 'Michael Brown', Position: 'Software Engineer' } ]
corporate_hub> db.employees.updateOne(
... {EmployeeID: 2001}, {$mul: {Salary: 1.1}}
...
corporate_hub> db.employees.updateOne(
... {EmployeeID: 2001}, {$mul: {Salary: 0.1}}
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
corporate_hub> db.employees.updateOne( { EmployeeID: 2001 }, { $set: { Department: "IT" } } )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
corporate_hub> show collections
employees
corporate_hub> db.employees.drop()
true
corporate_hub>

```

Activate Windows

Go to Settings to activate Windows.

3)

```
corporate_hub> use event_manager
switched to db event_manager
event_manager>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
event_manager> db.events.insertMany([
...   {
...     EventName: "Tech Conference 2024",
...     Location: "Colombo",
...     Date: ISODate("2024-12-15"),
...     Participants: 250
...   },
...   {
...     EventName: "Art Exhibition",
...     Location: "Kandy",
...     Date: ISODate("2024-12-20"),
...     Participants: 100
...   },
...   {
...     EventName: "Music Fest",
...     Location: "Galle",
...     Date: ISODate("2024-12-22"),
...     Participants: 500
...   }
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('677369c713de0e7bb72710bd'),
    '1': ObjectId('677369c713de0e7bb72710be'),
    '2': ObjectId('677369c713de0e7bb72710bf')
  }
}
```

```

event_manager> db.events.find({ Participants: { $gt: 200 } })
[
  {
    _id: ObjectId('677369c713de0e7bb72710bd'),
    EventName: 'Tech Conference 2024',
    Location: 'Colombo',
    Date: ISODate('2024-12-15T00:00:00.000Z'),
    Participants: 250
  },
  {
    _id: ObjectId('677369c713de0e7bb72710bf'),
    EventName: 'Music Fest',
    Location: 'Galle',
    Date: ISODate('2024-12-22T00:00:00.000Z'),
    Participants: 500
  }
]
event_manager> db.events.updateOne(
...   { EventName: "Art Exhibition" },
...   { $set: { Location: "Matara" } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
event_manager> db.events.deleteOne({ EventName: "Music Fest" })
{ acknowledged: true, deletedCount: 1 }
event_manager> show collections
events
event_manager> db.events.drop()
true
event_manager>

```

4)

```

event_manager> use hospital_db
switched to db hospital_db
hospital_db> db.createCollection("doctors")
{ ok: 1 }
hospital_db> db.createCollection("patients")
{ ok: 1 }
hospital_db> db.doctors.insertMany([
...   { "DoctorID": 1, "Name": "Dr. Smith", "Specialization": "Cardiology", "AvailableDays": ["Monday", "Wednesday"], "Patients": [] },
...   { "DoctorID": 2, "Name": "Dr. Lee", "Specialization": "Pediatrics", "AvailableDays": ["Tuesday", "Thursday"], "Patients": [] }
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67739ce113de0e7bb72710c0'),
    '1': ObjectId('67739ce113de0e7bb72710c1')
  }
}
hospital_db> db.patients.insertMany([
...   { "PatientID": 101, "Name": "Alice Brown", "Age": 45, "Illness": "Heart Disease", "AssignedDoctorID": 1 },
...   { "PatientID": 102, "Name": "Michael Green", "Age": 8, "Illness": "Flu", "AssignedDoctorID": 2 }
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('67739d0213de0e7bb72710c2'),
    '1': ObjectId('67739d0213de0e7bb72710c3')
  }
}
hospital_db> db.patients.insertOne({
...   "PatientID": 103,
hospital_db>
...   "Age": 50,
...   "Illness": "High Blood Pressure",
...   "AssignedDoctorID": 1
... })
{
  acknowledged: true,
  insertedId: ObjectId('67739d2113de0e7bb72710c4')
}

```

```

...     PatientID: 103,
hospital_db>
...     "Age": 50,
...     "Illness": "High Blood Pressure",
...     "AssignedDoctorID": 1
... })
{
  acknowledged: true,
  insertedId: ObjectId('67739d2113de0e7bb72710c4')
}
hospital_db> db.doctors.updateOne(
...   { "DoctorID": 1 },
...   { $set: { "Patients": [101, 103] } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
hospital_db>

hospital_db> db.doctors.updateOne(
...   { "DoctorID": 2 },
...   { $set: { "Patients": [102] } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
hospital_db> db.patients.find({ "AssignedDoctorID": 1 })
[
  {
    _id: ObjectId('67739d0213de0e7bb72710c2'),
    PatientID: 101,
    Name: 'Alice Brown',
    Age: 45,
    Illness: 'Heart Disease',
    AssignedDoctorID: 1
  },

```



```

{
  _id: ObjectId('67739d2113de0e7bb72710c4'),
  PatientID: 103,
  Name: 'John Doe',
  Age: 50,
  Illness: 'High Blood Pressure',
  AssignedDoctorID: 1
}
]
hospital_db> db.inventory.insertOne({
...   "ItemName": "Surgical Masks",
...   "Quantity": 500,
...   "LastRestocked": "2024-12-01"
... })
{
  acknowledged: true,
  insertedId: ObjectId('67739ee613de0e7bb72710c5')
}
hospital_db> db.inventory.updateOne(
...   { "ItemName": "Surgical Masks" },
...   { $inc: { "Quantity": -100 } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
hospital_db> db.patients.deleteOne({ "PatientID": 102 })
{ acknowledged: true, deletedCount: 1 }
hospital_db> db.inventory.drop()
true
hospital_db>

```